CP000698.1:1646187-1656577
Geotalea uraniireducens Rf4

PIN MazE_antitoxin ParE_toxin FRG DUF1353 SecB MqsA_antitoxin MqsR_toxin Bac_DnaA Hda_lid RuvB_N AAA_lid_4 RuvB_C RuvA_N HHH_5 RuvA_C RuvC TACO1_YebC_N Transcrip_reg YbjQ_1 IclR_C HTH_IclR

AATVCI010000002.1:292025-300996
Salmonella enterica

DUF3487 DUF2976 DUF3262 Phage_ASH SymE_toxin Plasmid_RAQPRD MqsA_antitoxin MqsR_toxin DUF4400 ? TrwB_AAD_bind TraG-D_C DUF2859 ? ? ?

AMFJ01027514.1:59-762
uncultured bacterium

MqsA_antitoxin MqsR_toxin

ANIK01000056.1:31991-45312
Leptospira alstonii serovar Sichuan str. 79601

HD Receiver ? MqsR_toxin Hist_deacetyl MqsA_antitoxin ? SpoIIE DNA_pol_A_exo1 ? ? PK PK_C ? ChlI AAA_25 Zn_ribbon_LapB Melibiase

DF820468.1:303681-319510
Candidatus Vecturithrix granuli

Fer4_11 Fer4_9 ThiF ThiS Pan_kinase ? Zn_ribbon_4 REase ResIII Helicase_C EcoEI_R_C MqsR_toxin MqsA_antitoxin Methylase_S Methylase_S ? HNH HsdM_N N6_Mtase TED PEP-CTERM MFS_1

BDTC01000056.1:17632-33293
bacterium BMS3Abin10

TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C ILVD_EDD_N ILV_EDD_C DUF465 Radical_SAM TatD_DNase PseudoU_synth_1 PseudoU_synth_1 SecB MqsA_antitoxin MqsR_toxin ? HTH_3 Peptidase_M78 Mg_chelatase_C Mg_chelatase ChlI PAS HD_5 PAS PAS Aminotran_4 PCMT

BNUA01000061.1:554-9597
Betaproteobacteria bacterium

MqsA_antitoxin MqsR_toxin dnstrm_HI1420 Gp49 ? Na_H_Exchanger TrkA_N Receiver NtrC-AAA AAA_lid_14 HTH_8 PAS His-Kinase His-Kinase DUF4124

BNYX01000100.1:995-2740
Betaproteobacteria bacterium

BrnA_antitoxin MqsA_antitoxin MqsR_toxin DUF3418

DADWXZ010000032.1:25057-34453
Enterobacter hormaechei subsp. xiangfangensis

? ? PALP ? ? ? Xre_MbcA_ParS_C MqsR_toxin MqsA_antitoxin MipZ ? ? Rep3_N ? ?

DASSCZ010000076.1:1-3273
Longimicrobiaceae bacterium

? MqsA_antitoxin MqsR_toxin DUF5916 DUF5916

DASTRX010000116.1:49625-15123
Polyangiaceae bacterium

TerC His-Kinase Bact_transglu_N Transglutaminase CSD Ribosomal_S30AE HTH_32 HTH_28 MqsR_toxin MqsA_antitoxin OmdA 5_nucleotid_C Calcineurin Cpn60_TCP1 2OG-FeII_Oxy_2 ? CbiA

DATAAG010000425.1:184-11443

Longimicrobium sp.

Flavin_Reduct Peptidase_M14 NYN Unstab_antitox Thiolase_C Thiolase_N GDC-P GDC-P GcvP2_C ? MqsR_toxin MqsA_antitoxin DUF2087 HTH_5

DATABT010000848.1:138-4417
Longimicrobium sp.

? ? MqsR_toxin MqsA_antitoxin ? Peptidase_M20

DASUPY010000343.1:141-2789
Longimicrobium sp.

Peptidase_M20 MqsA_antitoxin MqsR_toxin

DROK01000052.1:2226-3396
Thermodesulfatator atlanticus

CN_hydrolase MqsR_toxin MqsA_antitoxin

DQTM01000189.1:1-7720
Sulfurospirillum arcachonense

DUF2971 MqsR_toxin MqsA_antitoxin ? ParE_like His-Kinase His-Kinase PAS PilJ Receiver Trans_reg_C nos_propeller Cupredoxin_1 Cytochrom_C

DATEQZ010000013.1:18369-34590
Candidatus Binatus sp.

Transket_pyr Transketolase_C Biotin_lipoyl E3_binding 2-oxoacid_dh GST_N_2 GST_C Receiver Receiver His-Kinase His-Kinase PAS Receiver Receiver His-Kinase His-Kinase PAS CHASE3 Bestrophin-like MqsR_toxin MqsA_antitoxin TetR_N TetR_C_33 Bac_luciferase Abhydrolase_1 ? GST_N_2 GST_C GST_N_2 GST_C

DATHBK010000033.1:29-8068
Longimicrobium sp.

Excalibur 3-dmu-9_3-mt Cupin_2 ? PadR FtsX MacB_PCD FtsX MacB_PCD MqsA_antitoxin MqsR_toxin Glyoxalase ? Peptidase_M24

DATIJJ010000078.1:13567-6972
Gammaproteobacteria bacterium

Methyltransf_31 MqsA_antitoxin MqsR_toxin ? NAGPA Pro_CA MatE MatE Adenine_glyco GIY-YIG Dabb

DATJOJ010000022.1:45833-61930
Terriglobales bacterium

? OMP_b-brl_4 Plug CarboxypepD_reg ? DUF5659 HTH_36 ? ? MqsR_toxin MqsA_antitoxin Leu_zip ? ? ? ? ?

DATNIV010000444.1:1-3216
Telmatospirillum sp.

MqsA_antitoxin MqsR_toxin ? HTH_17 ?

DASQBY010001245.1:211-11695
Pseudomonas sp.

Fer4_ETF_QO ETFQO_UQ-bd NAD_binding_8 MqsA_antitoxin MqsR_toxin MFS_1 Receiver Sigma70_r2 Sigma70_r4_2 HlyD_D23 FtsX MacB_PCD ABC_tran OEP OEP ?

DASPZC010000154.1:3123-12438
Xanthobacteraceae bacterium

PIN MazE_antitoxin MFS_1 MFS_1 Abhydrolase_3 Abhydrolase_6 Bac_luciferase MqsA_antitoxin MqsR_toxin FAA_hydrolase OMP_b-brl ?

JSZA02000028.1:128-10238

Candidatus Thiomargarita nelsonii

? Pur_ac_phosph_N Calcineurin Amidase_2 ATPase_2 TPRs TPRs ACBP4_C MqsR_toxin ? MqsA_antitoxin Peptidase_S9_N CN_hydrolase

JQIT01000002.1:422543-436892
Sulfuricurvum sp. PC08-66

? PIN ? HypD HupF_HypC cobW ? MqsR_toxin HycI HycI HypF_C_2 HypF_C Sua5_yciO_yrdC zf-HYPF zf-HYPF Acylphosphatase ? Ni_hydr_CYTB NiFeSe_Hases NiFe_hyd_SSU_C Oxidored_q6 TAT_signal

LMLI01000006.1:17615-26741
Serratia sp. Leaf50

? ? ? ? Antirestrict ? ? MqsR_toxin MqsA_antitoxin ? DUF5983 DUF932 ? NikA-like Relaxase TraI_middle LPD7

JACDBT010000440.1:118-14278
Gemmatimonadetes bacterium

MatE MatE ? Sigma70_r2 Sigma70_r4_2 zf-HC2 DUF4097 CBM9_1 DUF5916 DUF4097 MqsR_toxin MqsA_antitoxin ? Glyoxalase_3 UPF0203 Receiver PAS GAF_2 His-Kinase His-Kinase

JACUTW010000029.1:193-16447
Desulfuromonadales bacterium

TPRs TPRs Alkyl_sulf_dimr TPRs TPRs TPR_CcmH_CycH Guanylate_cyc HAMP sCache_3_2 NfeD NfeD_membrane NfeD1b_N Band-7 Acetylase MqsA_antitoxin MqsR_toxin HSP90 HATPase_c_3 NUDIX P-II Gln-synt_C Gln-synt_N Rab11-FIP3 OmpA QRPTase_C QRPTase_N BPL_C BPL_LplA_LipB HTH_11

JADGBP010000257.1:1219-4064
Desulfamplus sp.

VWA ? MqsR_toxin MqsA_antitoxin BatD BatD BatD

JADQDT010000011.1:69317-76920
Pseudomonas syringae pv. tomato

DUF2726 ? ? ? ? ? ? MqsR_toxin MqsA_antitoxin ? ? ? VRR_NUC Zn_ribbon_SprT GntR

WGMK01000008.1:23861-51242
Candidatus Hydrogenedens sp.

CCP_MauG ? Bact_surface_Ig-like CarboxypepD_reg Cohesin ABC2_membrane_3 ABC_tran HlyD_3 Biotin_lipoyl_2 PKD_4 MqsR_toxin MqsA_antitoxin HTH_17 YNCE Phosphoesterase Bact_surface_Ig-like Bact_surface_Ig-like K319L-like_PKD K319L-like_PKD PKD_4 Alginate_exp Pyridoxal_deC FGGY_C FGGY_N 3keto-disac_hyd

WGMK01000010.1:30778-47096

Candidatus Hydrogenedens sp.

YdjC ECH_1 ? VWA AIRS_C AIRS Carb_kinase SIS_2 SIS MqsR_toxin MqsA_antitoxin ? Ldh_1_N Ldh_1_C Reprolysin_5 TNFR_ELAPOR1_6th TNFR_ELAPOR1_6th TNFR_ELAPOR1_6th TNFR_ELAPOR1_6th Pribosyltran_N Pribosyl_synth Ribosomal_L25p Ribosomal_TL5_C Pept_tRNA_hydro

JACPGD010000371.1:14999-23798

Candidatus Hydrogenedentota bacterium

Rrf2 B12-binding Radical_SAM ApbA ApbA_C ? Pro_isomerase Pro_isomerase ? MqsR_toxin MqsA_antitoxin HTH_17 ATP-synt_ab Rho_RNA_bind Rho_N DNA_pol_A

JACQBS010000161.1:1-7563
Nitrospinota bacterium

DDE_Tnp_ISL3 ? Radical_SAM SPASM PCP_red MqsR_toxin MqsA_antitoxin SecB ? ADH_zinc_N ADH_N DeoC AAA AAA_assoc_2 MgsA_C ORF6N

JACQEK010000096.1:3895-18550
Deltaproteobacteria bacterium

Bac_luciferase ABC_tran ABC_tran CbiQ Rieske Dimerisation Methyltransf_2 Glyco_transf_4 Glycos_transf_1 MqsA_antitoxin MqsR_toxin Guanylate_cyc TPRs TPRs TPRs TPRs TPRs TPRs TPRs ? ? TPRs TPRs Guanylate_cyc Acetylase Abhydrolase_1

JACRAQ010000078.1:14381-27247
Nitrosomonadales bacterium

MoCF_biosynth GST_C_2 GST_N_2 cNMPBD CBS CBS DUF294 DUF294_C RNase_T PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C FAD_binding_4 FAD-oxidase_C NYN OST-HTH MqsR_toxin MqsA_antitoxin ? Arg_tRNA_synt_N tRNA-synt_1d DALR_1

JACRCE010000018.1:394-3625
Deltaproteobacteria bacterium

MqsR_toxin MqsA_antitoxin SecB DsrE NIR_SIR NIR_SIR_ferr Fer2_BFD Fer2_BFD

JACRNB010000016.1:182913-192347
Betaproteobacteria bacterium

CobD_Cbib WCX WYL HTH_11 Pyr_redox_2 FCSD_central FCSD-flav_bind Cytochrome_CBB3 Cytochrom_C Ni_hydr_CYTB Cytochrome_C7 Paired_CXXCH_1 Mtrc-MtrF_II-IV_dom Gate Gate MqsA_antitoxin MqsR_toxin Beta-Roll Beta-Roll Beta-Roll

JAEHTF010000005.1:131966-148774
Pseudomonas putida

? OEP OEP FtsX MacB_PCD ABC_tran HlyD_D23 Sigma70_r2 Sigma70_r4_2 Receiver MFS_1 MqsR_toxin MqsA_antitoxin ? Fer4_ETF_QO ETFQO_UQ-bd NAD_binding_8 ETF ETF ETF_alpha SBP_bac_3 DUF6491 DUF4398

NRRQ01000083.1:1-1820
Chromatium okenii

CbiA MqsR_toxin MqsA_antitoxin ? DnaB_C

JAENWA010000007.1:1738-15071
Desulfuromonadales bacterium

RecR_HhH RecR_ZnF Toprim_4 RecR_C YbaB_DNA_bd DNA_pol3_delta2 DNAX_ATPase_lid DNA_pol3_gamma3 YHYH PNPase Caroten_synth MqsA_antitoxin MqsR_toxin ResIII ? RE_endonuc RE_endonuc N6_N4_Mtase ? DUF5655 NucS_C

JADJBD010000004.1:75622-90768
Leptospiraceae bacterium

? ? rve rve_3 HTH_37 Gp49 MCPsignal PAS DUF4140 DUF4139 MqsA_antitoxin MqsR_toxin DUF1389 Receiver His-Kinase His-Kinase PAS PAS PAS PAS PAS Uma2 ? Polbeta HEPN HEPN ?

JADJBD010000056.1:25729-40305
Leptospiraceae bacterium

DMTF1_N ADH_zinc_N ADH_N FUR PEGA Peptidase_M23 Pterin_bind GAF_2 GAF NtrC-AAA AAA_lid_14 HTH_8 SpoIIE_N MqsR_toxin NHL Vgb_lyase NHL ? DUF1554 DUF4114 DUF4842 ? MORN_2 MORN_2 Guanylate_cyc

JADJJA010000001.1:834421-849899
Candidatus Obscuribacter sp.

CobS TPRs TPRs TPRs Gly-zipper_YMGG ? GT87 Spermine_synth TPRs MqsR_toxin MqsA_antitoxin APH ? ? EamA EamA HTH_24 AsnC_trans_reg Polyketide_cyc2

JADJQZ010000001.1:5150399-5164237
Candidatus Obscuribacter sp.

Thymidylate_kin Tyr_Deacylase FRG ? Mu-transpos_C_2 ? MqsA_antitoxin MqsR_toxin ? ? ? DUF1963 DUF429 DUF1345 SEC-C

JADJWY010000001.1:266246-280236
Hahellaceae bacterium

? Ferritin Fer2_BFD VIT1 VIT1 DUF211 ? MqsA_antitoxin MqsR_toxin AA_kinase ACT_9 HI0933_like HTH_18 Arabinose_bd Patatin DUF2505 Amidohydro_1 DUF6164

JADKJU010000003.1:510927-519419
Betaproteobacteria bacterium

UvrD_C_2 PIN ? MarR_2 NTP_transf_2 ? Zn_ribbon_Top1 Mrr_cat UvrD_C_2 MqsR_toxin MqsA_antitoxin ORF6N UvrD_C_2 dnstrm_HI1420 AbiEi_2 ? ?

JAEUWI010000046.1:27649-51131

Candidatus Hydrogenedentota bacterium

Glycos_transf_1 Glyco_transf_4 PQQ_2 PQQ_2 Beta-prop_ACSF4 ABC_tran FtsX FtsX ? Beta_helix Beta_helix MqsA_antitoxin HTH_17 MqsR_toxin DUF4838 Glyco_hydro_2_N AXE1 AXE1 Peptidase_M56 Penicillinase_R TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C UDPGP

JAEUJJ010000003.1:11683-29882
Gemmatimonadetes bacterium

Lactamase_B_2 Amidase PIN RHH_1 ResIII SecB MqsA_antitoxin MqsR_toxin N6_N4_Mtase Phasin MotA_ExbB OmpA HNH_3 NERD AAA_19 UvrD_C_2 ? ?

VGZQ01000136.1:1-4114
Rhodospirillales bacterium

? DHDPS SmpB MqsA_antitoxin MqsR_toxin Guanylate_cyc

JAFFPR010000051.1:6301-11273
Leptospira abararensis

? PIN PhdYeFM_antitox HicB HicA_toxin ? MqsA_antitoxin MqsR_toxin

JAGCPQ010000181.1:496-4721
Neisseriaceae bacterium

MqsA_antitoxin MqsR_toxin HicB_lk_antitox HicB-like_2 HicA_toxin ? ?

JAGNGZ010000039.1:25111-41005
bacterium

CTP_transf_like HrpE Beta-lactamase2 SpoIIM RDD Citrate_synt Eco57I MqsR_toxin MqsA_antitoxin ? Tmemb_cc2 LRR_14 STY-kinase Acetylase HIT ?

JAFWGA010000014.1:1689-17341
Blautia sp.

CRISPR_assoc Cas_Cas5d Cas_CT1975 CRISPR_Cse2 CRISPR_Cse1 Cas3_HD DEAD Cas3-like_C_2 MqsA_antitoxin SocA_Panacea MqsR_toxin SBP_bac_5 ABC_tran ABC_tran oligo_HPY BPD_transp_1 BPD_transp_1_N BPD_transp_1 DUF6290 ParE_toxin

JAFZLA010000002.1:38133-53184
Treponema sp.

HTH_3 HipA_C Couple_hipA TPT EamA GumK_N Glyco_trans_1_4 UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C MqsA_antitoxin SocA_Panacea MqsR_toxin Glycos_transf_2 GtrA_DPMS_TM Glycos_transf_2 ? Glycos_transf_2 Glycos_transf_1 DUF6492 ?

JAFAMT010000289.1:100-4734
Herbaspirillum sp.

IDH MqsR_toxin MqsA_antitoxin MFS_1 Sugar_tr

JAFDHL010000100.1:15629-9371
Deltaproteobacteria bacterium

OMP_b-brl ? SecB MqsA_antitoxin MqsR_toxin His-Kinase His-Kinase PAS Ribosomal_L21p Ribosomal_L27 RsgA_GTPase MMR_HSR1 GTP1_OBG PUA AA_kinase Aldedh CTP_transf_like

JAIBAQ010000049.1:22819-35300
Candidatus Hydrogenedentota bacterium

Esterase_PHB tRNA_Me_trans_C tRNA_Me_trans_M tRNA_Me_trans Bactofilin Peptidase_M23 Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N ? SQS_PSY MqsR_toxin MqsA_antitoxin HTH_17 ?

JAIEQI010000096.1:1-11086
Candidatus Obscuribacterales bacterium

? ? Hydant_A_N Hydantoinase_A Hydant_A_C Hydantoinase_B Hydantoinase_B MqsR_toxin MqsA_antitoxin Polyketide_cyc2 Acetylase HIT ATG22 Abhydrolase_2 CoA_transf_3

JAIENF010000059.1:321-4650
Burkholderiales bacterium

? Gp49 HTH_37 HicB_lk_antitox DUF4926 HTH_31 Unstab_antitox MqsR_toxin MqsA_antitoxin VWA

JAIUOX010000112.1:1-15429
Proteobacteria bacterium

Wzy_C Polysacc_synt Glycos_transf_1 Glyco_transf_4 His-Kinase Receiver SpoIIE Glyco_trans_1_4 Glyco_transf_4 Epimerase MqsR_toxin MqsA_antitoxin ? SLBB SLBB_2 SLBB SLBB SLBB SLBB Poly_export ? STAS_2 Glyco_transf_4 Glyco_trans_1_4 Glyco_tranf_2_3 Bac_transf

JAGRNI010000022.1:39861-53715
Leptospiraceae bacterium

? ZnR FHA Lipase_chap Abhydrolase_1 ACR_tran ? Uma2 MqsR_toxin MqsA_antitoxin HTH_17 ? Beta-lactamase ? ? ? SLT

JAGRIQ010000504.1:1-2512
Pseudomonadales bacterium

MoeA_C MoCF_biosynth MqsA_antitoxin MqsR_toxin PIN ? ?

JADLES010000026.1:5230-19155
Burkholderiales bacterium

Aminotran_5 DNA_topoisoIV DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C Amidohydro_1 OmpA Methyltransf_23 HAD_2 MqsA_antitoxin MqsR_toxin ? HTH_Crp_2 cNMPBD DUF2249 COX15-CtaA ? COX2 COX1

JADLGX010000063.1:1-3468
Candidatus Eisenbacteria bacterium

NtrC-AAA AAA_lid_14 HTH_8 MqsR_toxin MqsA_antitoxin SecB ?

JADZAO010000068.1:1-5586
Rhodobacteraceae bacterium

HTH_3 ? ? CbiA MqsR_toxin MqsA_antitoxin KfrA_N AAA

JADZBU010000215.1:12225-10855
Chromatiaceae bacterium

DUF4342 ? LptF_LptG MqsR_toxin MqsA_antitoxin F_bP_aldolase PK_C PK PGK Gp_dh_C Gp_dh_N Transketolase_C_1 Transket_pyr Transketolase_N AhpC-TSA

JAGGTR010000125.1:17003-11158
Deltaproteobacteria bacterium

TPRs TPRs Rod_CreS OmpA PD40 PD40 TolB_N TonB_2 ExbD MotA_ExbB DnaJ MqsR_toxin MqsA_antitoxin ? MerR_1 ? ? DnaB_C DnaB_C ?

JAGLTQ010000131.1:13314-19399
Gammaproteobacteria bacterium

? GST_C_2 GST_N_3 ANKs ? ? ? MqsA_antitoxin MqsR_toxin AbiEii AbiEi_3

JAKLLO010000005.1:35875-54041
Zoogloea sp.

Zn_ribbon_recom Recombinase Resolvase rve ? AAA_22 ? ? MqsA_antitoxin MqsR_toxin WYL WCX AAA AAA ? ? ? AAA_27 SMC_N Calcineurin

JALNZB010000029.1:4321-17500
Sulfuritalea sp.

TPRs TPRs OsmC COQ7 CNP1 NUDIX tRNA-synt_2b HGTP_anticodon SLT MqsR_toxin MqsA_antitoxin Mrr_N SRP_SPB SRP54 SRP54_N Cytochrom_C_asm CNNM CBS CorC_HlyC T2SSF T2SSF A24_N_bact Peptidase_A24

WQUQ01000052.1:2592-7298
Desulfobulbus sp.

FliS FliD_N FliD_C FlaG Flagellin_N Flagellin_C ParE_toxin PhdYeFM_antitox MqsA_antitoxin MqsR_toxin

JAHDYK010000070.1:374-9490
Burkholderiales bacterium

? MqsA_antitoxin MqsR_toxin HNH_2 tRNA-synt_1c tRNA-synt_1c_C tRNA-synt_1c_C2 MazE_antitoxin PIN WS_DGAT_C WS_DGAT_cat Palm_thioest Abhydrolase_1

JAAEMW010001765.1:615-8591
Desulfobacteraceae bacterium

dTDP_sugar_isom RmlD_sub_bind MqsR_toxin MqsA_antitoxin tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c ALGX ALGX MBOAT

JANLIK010000021.1:208-1939
archaeon

MqsR_toxin MqsA_antitoxin SecB ?

JAPLKX010000644.1:2838-9875
Candidatus Hydrogenedentota bacterium

Glyco_hydro_106 DEAD Helicase_C MqsR_toxin MqsA_antitoxin HTH_17 HGD-D HGD-D MarR MFS_1 Acyltransferase AMP-binding

JAQIBN010000101.1:585-6466
Spirochaetia bacterium

? MqsR_toxin MqsA_antitoxin ? SLFN_AlbA_2 HTH_3 ? ParD_antitoxin DUF772 DDE_Tnp_1_6

JAQBQD010000041.1:30232-50776
Gemmatimonadetes bacterium

Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10 ? Glyco_trans_1_4 Glyco_transf_4 Peptidase_M23 ? GatB_Yqey tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c ? MqsR_toxin MqsA_antitoxin DcuC ? ? APH MTHFR S-methyl_trans ?

JAQUAG010000002.1:28580-43150
Arcobacteraceae bacterium

CZB MCPsignal HAMP dCache_3 PAS GGDEF PAS HK_sensor_dom_bact ? His-Kinase Redoxin MqsA_antitoxin MqsR_toxin Phage_int_SAM_3 Phage_integrase ? Prominin OEP OEP HlyD_D23 ABC_tran FtsX MacB_PCD

JAQTOA010000019.1:37445-47172
Sterolibacterium sp.

? ? DnaB DnaB_C HTH_36 ? ? MqsA_antitoxin MqsR_toxin ? Peptidase_S24 ? ? ? Arm-DNA-bind_3 Phage_integrase HTH_1

JAQTMZ010000005.1:157823-171628
Gallionella sp.

CNNM CBS CorC_HlyC ? Na_H_Exchanger Cation_efflux Ub-RnfH nSTAND_NTPase4 Calcineurin MqsA_antitoxin MqsR_toxin Toxin_YhaV PrlF_antitoxin Ub-RnfH Rnf-Nqr FMN_bind NQR2_RnfD_RnfE RnfC_N Complex1_51K SLBB Fer4_7

JAOUEW010000001.1:542162-555574
Deltaproteobacteria bacterium

SNase ? ? Cytochrom_C_2 ? TPRs ? MqsR_toxin MqsA_antitoxin ? Fic DUF977 AAA_15 OLD-like_TOPRIM UvrD-helicase UvrD-helicase UvrD_C ? PIN

JAJSZQ010000067.1:2837-12426
Deltaproteobacteria bacterium

? ? ? ? PIN DUF2281 RE_endonuc ResIII MqsR_toxin MqsA_antitoxin SocA_Panacea ?

JAJDIM010000019.1:10208-18266
Desulfovibrio sp. OttesenSCG-928-F20

ABC_tran oligo_HPY BFN_dom UPF0004 Radical_SAM TRAM Bac_DNA_binding TPRs ? MqsA_antitoxin MqsR_toxin Fic_N Fic

JAUCGE010000031.1:15752-31743
Desulfobacterales bacterium HSG16

STAS dCache_2 MCPsignal ANTAR GAF DUF362 FMN_red ResIII RE_endonuc SLFN_AlbA_2 HATPase_c_4 MqsR_toxin MqsA_antitoxin VapB_antitoxin ? PIN MoxR-AAA McrBC ?

JAUSKU010000081.1:25483-41522
Deltaproteobacteria bacterium

PIN SBP_bac_11 PHP_C Na_Ca_ex Na_Ca_ex Pyr_redox_2 N6_N4_Mtase ResIII MqsR_toxin MqsA_antitoxin SecB NAD_Gly3P_dh_N NAD_Gly3P_dh_C Sua5_yciO_yrdC ChlI Mg_chelatase Mg_chelatase_C MS_channel_2nd MS_channel_3rd cNMPBD

JAUXWC010000046.1:24393-36176
Rhodoferax sp.

? PemK_toxin AbiEi_1 ? RHH_1 PIN MqsA_antitoxin MqsR_toxin PIN FitA-like_RHH ? Secretin_N Secretin_N Secretin ? ? ?

JAUYFB010000027.1:16916-32721
Rectinemataceae bacterium

HTH_18 Lactamase_B Beta-Casp RMMBL HTH_18 GGDEF_2 DctP DctQ DctM DUF2264 MqsR_toxin WG WG WG WG WG ABC_tran LptD_N LptC

JAUPKZ010000001.1:1-9949
Campylobacterota bacterium

MqsA_antitoxin MqsR_toxin H2O2_YaaD Pribosyl_synth Pribosyltran_N LepA_C EFG_C RF3_C GTP_EFTU_D2 GTP_EFTU Pribosyltran ? ? EAL GGDEF

JAUPLR010000028.1:12576-23610
Campylobacterota bacterium

UPF0547 MscL ? Thioredoxin_7 ? Thioredoxin_7 Polbeta NTase_sub_bind MqsR_toxin MqsA_antitoxin ResIII UvrB_inter Helicase_C UvrB UVR VTT_dom DUF808 Transpeptidase Transgly ABC_tran Aminotran_1_2

JAVHLK010000173.1:4872-6751
Desulfomonilaceae bacterium

REase ? ABC-ATPase AAA_15 ? MqsA_antitoxin MqsR_toxin

JAISXB010000085.1:14477-10352
Azoarcus sp.

? MqsR_toxin MqsA_antitoxin PAS PAS PAS PAS His-Kinase His-Kinase Receiver Hpt Receiver GGDEF PP-binding AMP-binding AMP-binding_C Orn_Arg_deC_N Acyltransferase

JAISTW010000027.1:57199-85378
Desulfovibrio sp.

TPRs Fer4_13 TPP_enzyme_C POR_N POR Peptidase_S11 DNA_pol_A 5_3_exonuc 5_3_exonuc_N FMN_dh FMN_dh MqsA_antitoxin MqsR_toxin Beta-Roll Big CADHERIN Big CADHERIN Cadherin ? PhoH HD 7TM-7TMR_HD 7TMR-HDED YbeY BPL_LplA_LipB Zn_Ribbon_1 Biotin_lipoyl PYC_OADA HMGL-like Biotin_carb_C CPSase_L_D2 Biotin_carb_N

JAISPJ010000012.1:1-6782
Azoarcus sp.

CagX Unstab_antitox MqsA_antitoxin MqsR_toxin ? ? KORA HicB DUF3617 Pyr_redox_2 Pyr_redox_dim PBP_like_2

JAITWK010000009.1:4921-22218
Deltaproteobacteria bacterium

BPD_transp_1 dCache_1 His-Kinase His-Kinase Receiver Peripla_BP_6 BPD_transp_2 BPD_transp_2 BPD_transp_2 ABC_tran BCA_ABC_TP_C ABC_tran BCA_ABC_TP_C MqsR_toxin MqsA_antitoxin CheW His-Kinase H-kinase_dim Hpt Hpt Receiver CheW ParA-Soj-NTPase CheR CheR_N HEATs HEATs HEATs HEATs HEATs HEATs

JAITWK010000031.1:12056-28492
Deltaproteobacteria bacterium

DBI_PRT Fer4_10 GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C Peptidase_S26 ChlI Mg_chelatase Mg_chelatase_C QueH Mntp MqsR_toxin MqsA_antitoxin CitMHS Bac_export_1 Bac_export_2 FHIPEP nSTAND_NTPase5 SRP54 ParA-Soj-NTPase ParA

JAWHJR010000004.1:262000-278110
Lachnospiraceae bacterium

Cas_Cas5d DUF3226 ABC-ATPase ? ? ? MqsA_antitoxin SocA_Panacea MqsR_toxin RicinB_lectin_2 Flg_new Flg_new Flg_new Flg_new DUF5411 ? ? ? ? T2SSE

JANRJL010000060.1:12046-26839
Rhodospirillaceae bacterium

DMRL_synthase GTP_cyclohydro2 DHBP_synthase Lum_binding Lum_binding RibD_C dCMP_cyt_deam_1 ATP-cone Zn_ribbon_NrdR SHMT LacAB_rpiB MqsR_toxin MqsA_antitoxin DUF6898 Methyltransf_11 Penicil_amidase MotB_plug OmpA ATE_N ATE_C DNA_gyraseA_C DNA_gyraseA_C DNA_topoisoIV

JAVEVG010000520.1:1-3765
Candidatus Vecturithrix sp.

EamA ? MqsR_toxin MqsA_antitoxin HEPN Polbeta Polbeta HEPN

JAVESU010000004.1:70512-80019
Desulfomicrobium sp.

? RecT DUF3799 ? Fic SecB MqsA_antitoxin MqsR_toxin HTH_3 Peptidase_S24 ? ? ? ? RusA ?

JAYEGS010000122.1:1-16927
Pseudomonadota bacterium

Alpha-amylase_C Alpha-amylase Alpha-amylase CBM_48 UvrD-helicase UvrD_C DEAD Helicase_C MZB SecB MqsA_antitoxin MqsR_toxin ParA-Soj-NTPase ? ABC-ATPase DUF3696 GmrSD_N ?

JAYEKV010000177.1:214-3482
Campylobacterota bacterium

HD HD_5 EamA EamA ? MqsR_toxin MqsA_antitoxin

WKKR01000001.1:131272-148403
Geobacter sp.

CDP-OH_P_transf AMP-binding AMP-binding_C PP-binding Acyltransferase Cation_ATPase_N E1-E2_ATPase Hydrolase Cation_ATPase_C HMA FtrD-like MacB_PCD FtsX ABC_tran MqsR_toxin MqsA_antitoxin DUF5677 ? Mrr_cat ? Y1_Tnp His-Kinase His-Kinase PAS PAS

SEJA01000061.1:2248-6685
Methylocaldum sp. BRCS4

DDE_Tnp_1 DUF4096 PhdYeFM_antitox PIN ? MqsA_antitoxin MqsR_toxin Xre_MbcA_ParS_C Xre-like-HTH DUF4359 DUF4359

VXPN01000252.1:396-2309

Boseongicola sp. SB0662_bin_57

MqsA_antitoxin MqsR_toxin GntR FCD

POWB01000033.1:1399-8984
Halochromatium sp.

Flp1_like ? Ank ANKs MqsR_toxin BolA Na_Ala_symp DUF3050 DsrC GATase_4 TrkA_N Na_H_Exchanger DUF1664

MFRL01000100.1:11317-8612

Candidatus Melainabacteria bacterium GWF2_37_15

GtrA_DPMS_TM Amino_oxidase Acetyltransf_6 MqsR_toxin MqsA_antitoxin ? RRM_1 Pantoate_transf Pantoate_ligase ? ?

MGSG01000010.1:2457-12227

Deltaproteobacteria bacterium RIFCSPLOWO2_02_FULL_53_8

? CRCB DUF190 TPRs TPRs SecB MqsA_antitoxin MqsR_toxin Sirohm_synth_M NAD_binding_7 Cytochrom_C_asm ? ParE_toxin Porphobil_deamC Porphobil_deam MotA_ExbB

MQVI01000008.1:42607-55355
Pasteurellaceae bacterium 15-036681

PmbA_TldD_3rd PmbA_TldD_2nd PmbA_TldD_1st DUF5339 PrmA DUF3800 ? HTH_3 MqsA_antitoxin MqsR_toxin HTH_3 HipA_C Couple_hipA Ser_hydrolase FGGY_N FGGY_C DctP DctM DctQ

QSLN01000015.1:23655-12348
Ammonifex thiophilus

? AAA_10 MqsA_antitoxin SocA_Panacea MqsR_toxin T2SSE ? TcpE RRXRR HNH TrbC bPH_2 TpcC

RBRO01000020.1:40267-55738
Pseudomonas savastanoi pv. glycinea

Receiver GerE HD_4 ? DUF6508 ? Synaptojanin ABC_tran AAA_assoc_C ? ? MqsR_toxin 5-nucleotidase His-Kinase His-Kinase ? SMI1_KNR4 BPD_transp_1 BPD_transp_1

SZXW01000003.1:26855-39334
Chlorobium sp.

Glycos_transf_1 Glyco_trans_4_4 Bac_transf LbH_EIF2B PglD_N DegT_DnrJ_EryC1 AbiEii ? MqsA_antitoxin MqsR_toxin HTH_3 AAA_14 DUF4143 HTH_3 Gp49 AAA_14 DUF4143 PhdYeFM_antitox ParE_toxin

SSEW01000052.1:5419-5290
Cellvibrionales bacterium

DUF2505 PALP Thr_dehydrat_C Thr_dehydrat_C MqsR_toxin MqsA_antitoxin ParD_like ParE-like_toxin Sel1 Sel1 Sel1 HTH_3 HigB-like_toxin Sod_Cu

CP119325.1:2436043-2450818
Candidatus Pseudomonas phytovorans

OEP OEP FtsX MacB_PCD ABC_tran HlyD_D23 Sigma70_r2 Sigma70_r4_2 Receiver MFS_1 MqsR_toxin MqsA_antitoxin Fer4_ETF_QO ETFQO_UQ-bd NAD_binding_8 ETF ETF ETF_alpha SBP_bac_3 DUF6491 DUF4398 DUF4398 OmpA Copper-bind

NZ_JAINUS010000208.1:14416-29239
Escherichia coli

Cu-oxidase_3 Cu-oxidase Cu-oxidase_2 Flavodoxin_2 Radical_SAM_N Radical_SAM DUF3362 GyrI-like MqsA_antitoxin His-Kinase His-Kinase 2CSK_N BOF LPP DUF2645 ABM Acyltransferase Trans_reg_C Receiver DNA_topoisoIV DNA_gyraseA_C DNA_gyraseA_C SBP_bac_5 MqsR_toxin

NZ_AHHC01000042.1:71054-86554
Pseudomonas sp. PAMC 25886

ABC_tran BPD_transp_1 DUF1120 PapD_N PapD_C DUF1120 DUF1120 MFS_1 FCD GntR MqsA_antitoxin BPD_transp_1 SBP_bac_3 Abhydrolase_2 PapC_N Usher PapC_C MqsR_toxin DEAD Helicase_C

NZ_CAUVFH010000154.1:1-2936
Kingella oralis

YdaS_toxin Peptidase_S24 HTH_3 TacA1 Acetyltransf_7 MqsR_toxin MqsA_antitoxin ?

NZ_SITF01000043.1:43509-59428
Hafnia paralvei

DAO DAO_C HTH_DeoR DeoRC Hexapep_GlmU NTP_transferase Glycos_transf_1 Glyco_transf_5 PapB Rhomboid_N Rhomboid Rhodanese GDPD BPD_transp_1 Phosphorylase ? GyrI-like HTH_18 MqsR_toxin ABC_tran OB_MalK TOBE_2 MqsA_antitoxin

NZ_CAWMAZ010000011.1:104065-118622
Yersinia kristensenii

BPD_transp_2 RelB-like HTH_3 ParE_toxin ABC_tran ABC_tran BPD_transp_2 ROK Fic_N Fic MotA_activ MqsA_antitoxin MqsR_toxin ? ? Peptidase_C80 DUF6979 DinI GerE

NZ_WHUG01000007.1:100929-112786
Rugamonas aquatica

Phasin_2 BPD_transp_2 DAGK_prokar IclR_C HTH_IclR Peptidase_S11 Nitroreductase EamA EamA MqsA_antitoxin MqsR_toxin ABC_tran BCA_ABC_TP_C ABC_tran BCA_ABC_TP_C BPD_transp_2 LYTB FKBP_C RadC UPF0758_N

NZ_ACJN02000001.1:1255117-1276720
Desulfonatronospira thiodismutans ASO3-1

ParE_like UvrD-helicase UvrD_C DEAD Helicase_C MZB HTH_3 MqsA_antitoxin MqsR_toxin N6_Mtase Eco57I Eco57I SNF2-rel_dom Helicase_C ? PIN HicB RF-1 ? ARG ARG ARG

NZ_WJYN01000001.1:1278939-1295175
Ralstonia pickettii

MqsA_antitoxin DUF4154 ATP-synt_DE_N ATP-synt_DE DAO Bac_luciferase Aldolase_II Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N CDO_I Rhodanese Rhodanese Rhodanese Rhodanese EamA EamA MqsR_toxin CHASE8 GGDEF EAL Plug TonB_dep_Rec_b-barrel LysR_substrate HTH_1 HAD

NZ_CAJPVI010000018.1:116110-129486
Cupriavidus numazuensis

MetW Lip_A_acyltrans DAP_epimerase DAP_epimerase PPK2 Pribosyltran Synaptojanin MqsA_antitoxin MqsR_toxin Semialdhyde_dhC_1 Semialdhyde_dh Peptidase_M48 MFS_1 4HBT_3 4HBT_3C EamA EamA LysR_substrate HTH_1 Patatin DUF3734

NC_007614.1:2500755-2517102
Nitrosospira multiformis ATCC 25196

HTH_29 rve rve_3 Glyco_transf_4 Methyltransf_25 DegT_DnrJ_EryC1 NYN OST-HTH BON BON MqsR_toxin MqsA_antitoxin ? Bax1-I tRNA_U5-meth_tr TRAM ? GFO_IDH_MocA GFO_IDH_MocA_C3 Receiver His-Kinase His-Kinase GAF_2 PAS PAS MqsA_antitoxin ?

NZ_CDRUZS010000005.1:243843-259611
Pseudomonas capeferrum

DUF6491 DUF4398 ETF ETF_alpha ETF MqsA_antitoxin Receiver Sigma70_r2 Sigma70_r4_2 MqsR_toxin FtsX MacB_PCD ABC_tran OEP OEP ? HlyD_D23 MFS_1 Fer4_ETF_QO ETFQO_UQ-bd NAD_binding_8 SBP_bac_3 DUF4398 OmpA

NZ_CGBP01000026.1:51095-6755
Yersinia similis

? DUF4942 MqsR_toxin MqsA_antitoxin ? ? NikA-like Relaxase TraI_middle LPD7 Antirestrict

NZ_CP158858.1:5199205-5214527
Herbaspirillum seropedicae

AHSA1 MFS_1 MFS_1 MFS_1 MFS_1 FCD GntR UTRA GntR ABC_tran MqsR_toxin BPD_transp_1 Glyoxalase EptA_B_N Sulfatase 4HB_MCP_1 HAMP MCPsignal Amidohydro_2 GST_C_4 GST_N_2 SBP_bac_3 BPD_transp_1

NC_018020.1:3562411-3578066
Turneriella parva DSM 21527

SH3_3 DUF7674 ? SocA_Panacea ? HTH_17 MqsR_toxin ? AHSA1 ? ? ? AHSA1 ? ?

NZ_KB892849.1:375-10077
Uliginosibacterium gangwonense DSM 18521

TIR MqsR_toxin MqsA_antitoxin HTH_1 LysR_substrate MCPsignal Mem_trans Acetyltransf_7 PurT_C ATP-grasp RS_preATP-grasp-like Beta-lactamase2

NZ_HG326655.1:127413-9529
Afipia birgiae 34632

Spermine_synt_N Spermine_synth MqsR_toxin MqsA_antitoxin ? ECH_1 AdoMet_dc HTH_Crp_2 cNMPBD PAS PAS GGDEF EAL DUF2336 ? EAL

NZ_ATUE01000005.1:499466-511682
Perlucidibaca piscinae DSM 21586

RNase_PH_C RNase_PH Peptidase_M24 Metal_hydrol MTS Synaptojanin Abhydrolase_1 MqsA_antitoxin MqsR_toxin Pirin Pirin_C MFS_1 Lipocalin_4 Pribosyltran GSH-S_ATP GSH-S_N TonB_C PrmA

NZ_JACVBG010000005.1:715303-732223
Pseudomonas sp. PDM19

DJ-1_PfpI HTH_18 OpuAC SDH_beta SDH_alpha OpuAC 3-dmu-9_3-mt SUKH_6 ? ? ? ? ? MqsA_antitoxin MqsR_toxin ? Bro-N

NZ_JAHRYT010000022.1:31079-45096
Acidovorax sp.

? ? Zn_ribbon_Nudix NUDIX TusA DUF4276 MqsA_antitoxin MqsR_toxin PALP ABC-ATPase Methylase_S Methylase_S HsdM_N N6_Mtase AAA_22 REase ResIII Helicase_C EcoEI_R_C DUF1090 DUF6279

NZ_JTET01000022.1:32735-41108
Hafnia alvei

MqsA_antitoxin HTH_DeoR DeoRC BPD_transp_1 ABC_tran OB_MalK TOBE_2 GDPD ? Rhodanese Rhomboid_N Rhomboid MqsR_toxin DAO DAO_C

NZ_AWNK01000005.1:262178-276511
Chrysiogenes arsenatis DSM 11915

PP-binding Fic Fic_N Radical_SAM B12-binding MqsR_toxin MqsA_antitoxin ? ? Radical_SAM B12-binding Radical_SAM B12-binding ABC_tran ABC2_membrane_3 ACP_syn_III_C Thiolase_N Lip_A_acyltrans LHD_TRPY1 DUF2062 Glycos_transf_2

NZ_BAYC01000029.1:11440-29765
Paraburkholderia fungorum NBRC 102489

DUF4136 CHASE2 His-Kinase His-Kinase FecR Receiver Trans_reg_C SWIM SNF2-rel_dom Helicase_C HTH_DeoR DeoRC MqsA_antitoxin MqsR_toxin GCV_T_C GCV_T FAO_M DAO Ni_hydr_CYTB Catalase DUF4148 ? Sigma70_r4_2 PhyR_sigma2 zf-HC2

NZ_PVEN01000009.1:259746-278028
Paraburkholderia sp. BL21I4N1

Sigma70_r4_2 PhyR_sigma2 DUF4148 zf-HC2 ? Catalase Ni_hydr_CYTB GCV_T_C GCV_T FAO_M DAO MqsR_toxin MqsA_antitoxin HTH_DeoR DeoRC SWIM SNF2-rel_dom Helicase_C Receiver Trans_reg_C FecR CHASE2 His-Kinase His-Kinase ?

NZ_DILG01000074.1:36593-48040
Sulfuricurvum sp. UBA5598

? ? Rotamase MqsA_antitoxin ? HATPase_c_3 HSP90 HSP90 LptF_LptG ? HTH_37 MqsR_toxin PseudoU_synth_1 PseudoU_synth_1 NnrU A24_N_bact Peptidase_A24 Prenyltransf Flavoprotein DFP DFP

NZ_CP034939.1:77302-89284
Pectobacterium odoriferum

MqsA_antitoxin ? ? ? ? DUF4942 ArdA MqsR_toxin ? ? ? ? ? ? PAPS_reduct

NZ_JAGGNZ010000005.1:1447-14948
Pseudomonas fluorescens

SnoaL_2 NA37 GIY-YIG Bac_DNA_binding Glutaredoxin GST_C_2 GST_N_3 MqsR_toxin MqsA_antitoxin ABC_tran BPD_transp_1 BPD_transp_1 SBP_bac_3 Abhydrolase_2 DEAD Helicase_C MFS_1

NZ_CP053748.1:50844-63671
Massilia sp. erpn

ClpS CSD Ribosomal_S20p Iso_dh ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N Citrate_synt HTH_17 MFS_3 Acetylase Acetylase MqsR_toxin MqsA_antitoxin PseudoU_synth_2 ? DUF192 MurJ

NZ_LMLM01000008.1:61331-74194
Duganella sp. Leaf61

Phage_integrase Phage_int_SAM_1 HTH_3 ? ? ? MqsR_toxin MqsA_antitoxin ? HTH_Tnp_1 TnpB_IS66 LZ_Tnp_IS66 zf-IS66 DDE_Tnp_IS66 DDE_Tnp_IS66_C cREC_REC ? Phage_integrase Receiver His-Kinase His-Kinase PAS PAS

NZ_LKHP01000017.1:11312-20899
Caloramator mitchellensis

2OG-FeII_Oxy_6 DUF5651 ? ? ? NYD-SP28_assoc HTH_7 MqsA_antitoxin SocA_Panacea Terminase_2 Terminase_3 Terminase_3C SPP1_portal Phage_min_cap2 Ntox50 ? SKA1 MqsR_toxin PDDEXK_15 ?

NZ_LKFB01000019.1:6524-16343
Comamonas thiooxydans

? HTH_36 ? MqsA_antitoxin MqsR_toxin ? ? ? DNA_ligase_A_M WGR CbiA ? ? ? ?

NZ_CBDKZX010000027.1:367-18262
Kluyvera sp.

GATase_7 Asn_synthase Asn_synthase Glucosamine_iso Hydrolase_6 Hydrolase_like MarR ROK NagA_N Amidohydro_1 PTS_EIIA_1 PTS_EIIB PTS_EIIC tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c OprD MFS_2 YbfN MqsR_toxin KdgM Glyco_hydro_67M Glyco_hydro_67C MqsA_antitoxin

NZ_CADFGN010000006.1:341757-353799
Paraburkholderia tropica

Tautomerase GFA Receiver Trans_reg_C MqsA_antitoxin MqsR_toxin MFS_3 MarR NUP Acetylase ABC_tran HAMP His-Kinase His-Kinase Methyltransf_25 DUF2071 ? LysR_substrate HTH_1

NZ_DAMDGG010000007.1:127047-140858
Enterobacter sp.

Na_Ca_ex Na_Ca_ex DAHP_synth_1 Transglut_core2 TPRs NIT ANTAR DsrE MCPsignal HAMP dCache_1 ChaC ChaB MqsA_antitoxin MqsR_toxin PrmC_N MTS PCRF RF-1 GlutR_N Shikimate_DH GlutR_dimer LolB GHMP_kinases_C GHMP_kinases_N

NZ_FRDI01000017.1:7549-19446
Desulfovibrio litoralis DSM 11393

MazE_antitoxin PemK_toxin HTH_17 MqsA_antitoxin MqsR_toxin Phage_integrase NOV_C AAA_23 T6SS_Tle1-like_cat Zn_ribbon_ISL3 DDE_Tnp_ISL3 Lcl_C Big Big IAT_beta ? MC2 CTD6

NZ_MUYA01000004.1:30638-42048
Haemophilus paracuniculus

Cytochrom_C_asm CorC_HlyC CBS CNNM PP-binding DUF406 YceG MqsA_antitoxin MqsR_toxin DNApol3-delta_C HolB_lid DNA_pol3_delta2 TatD_DNase DUF1523 ? Calcineurin Glycos_transf_2 Chlorophyllase Thymidylate_kin

NZ_LDOS01000001.1:586199-602964
Metallibacterium scheffleri

UbiD_C UbiD UbiD_N tyr_de_CO2_C Pyridoxal_deC Pyridoxal_deC CM_2 PDH_N PDH_C ? HTH_23 rve Mu-transpos_C_2 IstB_IS21 MqsA_antitoxin ? DDE_Tnp_1_6 DUF772 DDE_Tnp_1 MqsR_toxin ? ? ? TniQ

NZ_CP021425.1:201673-221500
Oleiphilus messinensis

DDE_Tnp_1 DUF772 DUF4291 ? SUKH_6 ? PPC DUF4214 DUF4214 MqsA_antitoxin MqsR_toxin HD DJ-1_PfpI HTH_18 Ring_hydroxyl_A Rieske Receiver Receiver His-Kinase Receiver EAL PAS PAS His-Kinase His-Kinase Receiver Hpt His-Kinase HAMP 4HB_MCP_1

NZ_FMXO01000029.1:163-3725
Desulfonatronum thiosulfatophilum

MqsA_antitoxin MqsR_toxin Sod_Fe_N Sod_Fe_C B12-binding Radical_SAM DUF4070 ?

NZ_PDNW01000015.1:38089-11074
Pollutimonas subterranea

ECH_1 EAL DUF1036 DUF4189 MqsR_toxin MqsA_antitoxin polyprenyl_synt DUF815 NUDIX TMP-TENI GGDEF GAF Cytochrom_C EAL

NZ_JAOCJT010000049.1:211-16203
Pseudomonas sp. GD03689

ETF DUF4398 OmpA DUF6491 DUF4398 SBP_bac_3 ETF ETF_alpha Fer4_ETF_QO ETFQO_UQ-bd Thi4 MqsA_antitoxin MqsR_toxin MFS_1 ? HlyD_D23 FtsX MacB_PCD ABC_tran OEP OEP ? Sigma70_r2 Sigma70_r4_2

NZ_PVLR01000002.1:2831-15342
Malikia spinosa

NfeD Band_7_C Band-7 Phage_integrase Phage_int_M Arm-DNA-bind_3 NA37 ? MqsA_antitoxin MqsR_toxin Flavin_Reduct Couple_hipA HipA_C ? ? SWI2_SNF2 ? DUF29 HTH_3

NZ_RQFT01000009.1:58132-71792
Leptospira bouyouniensis

DUF2001 ? ? ? ? ? ? MqsR_toxin MqsA_antitoxin ? ? MmeI_C MmeI_TRD MmeI_Mtase MmeI_hel MmeI_N ? ? ?

NZ_QDKJ01000017.1:35578-52385
Brenneria roseae subsp. americana

PDZ PDZ_2 Trypsin HD_assoc HD PNPase ? Fe-S_biosyn Aminotran_3 MqsR_toxin MqsA_antitoxin Aminotran_1_2 GntR LysE Peripla_BP_2 ABC_tran Transpeptidase Transgly UB2H PBP1_TM Peripla_BP_2 FecCD FecCD

NZ_QGDU01000009.1:80583-11813
Fibrobacter sp. UWR4

MqsA_antitoxin MqsR_toxin dnstrm_HI1420 ? DUF4832 RicinB_lectin_2 Glyco_hydro_16 tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C Gp49 ? Secretin

NZ_QTPP01000016.1:82662-99335
Burkholderia sp. Bp9142

Fer2_2 Fer2 PAS GGDEF ? AstE_AspA_cat SBP_bac_3 Cytochrom_C Cytochrom_C MoCoBD_2 MoCoBD_2 MoCoBD_1 MqsR_toxin MqsA_antitoxin Acetylase Bac_luciferase Bac_luciferase BPD_transp_1 ABC_tran TAT_signal SBP_bac_3 ?

NZ_SGXC01000001.1:1313800-1328005
Pigmentiphaga kullae

FAD_binding_3 BPD_transp_1 ABC_tran BPD_transp_1 SBP_bac_3 MqsA_antitoxin MqsR_toxin HmuY Plug TonB_dep_Rec_b-barrel TctC RraA-like HTH_IclR FAA_hydrolase LysR_substrate HTH_1 LysR_substrate HTH_1

NZ_SKBM01000059.1:9337-9002
Roseicella aquatilis

HTH_37 Peptidase_M78 ? ? ? GTP-bdg_N GTP-bdg_N GTP-bdg_M MMR_HSR1 HflX_C MqsR_toxin MqsA_antitoxin Cupin_2 RE_NgoFVII DUF4411 SNase

NZ_JBPDVQ010000003.1:93319-107392
Lonepinella koalarum

Sulfatase TadF malic Malic_M PTA_PTB MqsR_toxin TadE TPRs Tad MqsA_antitoxin TctC TctB TctA DHDPS PfkB MazE_antitoxin PemK_toxin

NZ_RQHT01000011.1:15138-24906
Leptospira bandrabouensis

? ? HTH_3 ? DUF5131 ? DUF1366 MqsR_toxin MqsA_antitoxin ParA-Soj-NTPase DUF3102 HTH_36 ? DUF4325 SRAP

NZ_JACHGA010000025.1:14327-22637
Rhizobium rosettiformans

HTH_3 HicB DUF6447 Phage_integrase MqsR_toxin DEDD_Tnp_IS110 Transposase_20 Beta-Roll MqsA_antitoxin HTH_Tnp_1 HTH_21 rve rve_3 HicB

NZ_CP041669.1:39059-54583
Legionella israelensis

? TrwC AAA_30 TraI_2B AAA_30 DUF7146 Toprim_3 Tra_M ? TraS ? ArdA MqsR_toxin MqsA_antitoxin ? Endonuclease_1 Peptidase_S24 IMS IMS_HHH_1 IMS_C DUF4113 rve_3 rve HTH_21 HTH_Tnp_1 GFA

NZ_JAOASZ010000053.1:7856-21597
Fluviibacter phosphoraccumulans

Pribosyltran OTCace_N OTCace Amidohydro_3 YggU P5CR_dimer F420_oxidored Ala_racemase_N T2SSE MqsR_toxin UPF0020 THUMP RlmL_1st UPF0093 Epimerase Fer2 FAD_binding_6 NAD_binding_1 DNA_pol_A DNA_pol_A_exo1 5_3_exonuc 5_3_exonuc_N DNA_processg_A Lysine_decarbox MqsA_antitoxin

NZ_JAACKB010000080.1:656-1386
Sulfurovum sp. bin170

MqsR_toxin MqsA_antitoxin

NZ_CAWPLU010000002.1:7014-16306
Candidatus Regiella insecticola

DDE_3 MqsR_toxin Phage_lysis DDE_Tnp_4 HTH_Tnp_4 ? Ogr_Delta Phage_lysozyme DDE_3 MqsA_antitoxin HTH_Tnp_IS630 PemK_toxin Phage_tail_3 HTH_Tnp_IS630 Phage_tail_S Phage_tail_S

NZ_BMXQ01000001.1:1955031-1972098
Pigmentiphaga litoralis

ABC_membrane ABC_tran Wzy_C 3-dmu-9_3-mt PHP DNA_pol3_alpha DNA_pol3_finger HHH_6 N-OB DNA_pol_IIIA_C MqsA_antitoxin MqsR_toxin adh_short_C2 ABC_tran ABC_tran BCA_ABC_TP_C BPD_transp_2 BPD_transp_2 Peripla_BP_6 Tautomerase Amidohydro_3 Amidohydro_3 Glyco_transf_9

NZ_NHSH01000077.1:19123-32026
Halochromatium roseum

Methyltransf_21 Methyltransf_21 DUF4351 ? ? ? MqsR_toxin MqsA_antitoxin Sulfotransfer_3 Glycos_transf_2 TylF RMNT_CmcI ? ? RMNT_CmcI PIN

NZ_DNES01000017.1:265754-282195
Hydrogenophaga sp.

Fer2 p450 Reductase_C Pyr_redox_2 HsdM_N N6_Mtase ? MqsA_antitoxin MqsR_toxin PIN7 DUF4357 SLATT_5 RVT_1 Methylase_S Methylase_S REase ResIII Helicase_C EcoEI_R_C ArsA_ATPase ArsA_ATPase PCC

NZ_JANZKU010000012.1:134752-143220
Pseudomonas sp. RIT-PI-AD

? ? Phage_TAC_7 Phage_P2_GpE PhageMin_Tail Phage_P2_GpU MqsR_toxin MqsA_antitoxin HicB-like_2 MethyltransfD12 ? Phage_GPD

NZ_JANBWQ010000055.1:24235-30702
Citrobacter portucalensis

? PerC HTH_23 ? ? ? ? MqsA_antitoxin MqsR_toxin HTH_31 ? ? ? ? ? ?

NZ_CP113061.1:1234899-1248556
Asticcacaulis sp. SL142

NAD_binding_10 Acetylase NUDIX ? Fer2 Hpt Arm-DNA-bind_3 Phage_int_M Phage_integrase MqsR_toxin MqsA_antitoxin HTH_17 EAL GGDEF Ion_trans TerC Zn_ribbon_TFIIB DUF4118 His-Kinase His-Kinase

NZ_CP115468.1:1374845-1391081
Leptospira sp. GIMC2001

Gp49 ? ? ? ? ? MqsA_antitoxin MqsR_toxin ? ? ? ? DUF2971 AbiJ_NTD4 DUF7014 HTH_CodY

NZ_CP115468.1:1777514-1790072
Leptospira sp. GIMC2001

MqsA_antitoxin MqsR_toxin PIN PhdYeFM_antitox DUF6678 ? FRG BamHI N6_N4_Mtase DUF5677 ? HTH_37 Gp49 DUF2971 ?

NZ_JAHCKZ010000002.1:16626-25316
Nitrosomonas sp.

? MqsR_toxin MqsA_antitoxin Peptidase_S24 DUF4113 IMS_C IMS ? DUF488 YjzC HTH_37 ? ? ? PDDEXK_1 ? QSregVF_b

NZ_AP027203.1:882532-914950
Polynucleobacter sp. SHI2

CTP_synth_N GATase DAHP_synth_1 Enolase_N Enolase_C DivIC HSP33 MFS_4 tRNA-synt_1c Anticodon_2 MqsR_toxin MqsA_antitoxin MORN MORN MORN MORN Gly_rich_2 Gly_rich_2 Gly_rich_2 Gly_rich_2 Gly_rich_2 Gly_rich_2 Gly_rich_2 TonB_C MotA_ExbB ExbD POTRA_2 ShlB

NZ_JARRAF010000034.1:8234-20576
Parachitinimonas caeni

Calcineurin GAAD ? Sel1 Sel1 HTH_Tnp_1 HTH_32 rve rve_3 ? ? MqsR_toxin Phage_int_SAM_1 Phage_integrase Phage_integrase Phage_integrase ? ? ? ? MqsA_antitoxin

NZ_CEVACK010000002.1:85186-100448
Desulfovibrio piger

? DUF6148 MqsA_antitoxin MqsR_toxin D5_N HTH_17 ? BRCT SUFBD_core Zn_ribbon_DnaG Toprim_2 ParB_N N6_N4_Mtase DUF1441 GpA_ATPase GpA_nuclease Phage_portal_2 Peptidase_S49

NZ_JAQTUH010000046.1:114-2729
Sulfuricurvum sp.

Rhodanese NTP_transf_2 MqsA_antitoxin MqsR_toxin EAL

NZ_JAUSMX010000005.1:206144-217459
Fluviicoccus sp.

Rieske Pyr_redox_2 Fer4 Molydop_binding Molybdopterin Molybdop_Fe4S4 HipA_C Couple_hipA HTH_31 TetR_N MqsR_toxin MqsA_antitoxin ? Lactamase_B B12-binding Radical_SAM

NZ_CABUWE010000327.1:1-3655
uncultured Bilophila sp.

Phage_portal_2 DUF6148 MqsA_antitoxin MqsR_toxin HTH_17 GpA_ATPase GpA_nuclease

NZ_JAHLDA010000091.1:16537-25999
Burkholderia gladioli

TrbC T2SSE HTH_3 RepA_C RepA_C ? ? ? MqsR_toxin MqsA_antitoxin DUF3717 ? ? HrpB1_HrpK ? ?

NZ_CP192775.1:541355-553744
Agrobacterium sp. rho-13.3

AHSA1 DUF1178 CN_hydrolase Glutaredoxin Pribosyltran DZR_2 Methyltransf_11 Flp_Fap MqsR_toxin MqsA_antitoxin Biotin_lipoyl Biotin_carb_C CPSase_L_D2 Biotin_carb_N CT_A_B CT_C_D D-Glu_cyclase LamB_YcsF Nramp GntR FCD

NZ_DATIND010000142.1:14123-29347
Longimicrobium sp.

XdhC_CoxI ELFV_dehydrog_N ELFV_dehydrog zf-dskA_traR ? GDC-P GDC-P GcvP2_C PIN PhdYeFM_antitox MqsR_toxin MqsA_antitoxin DUF4390 Omp85 Bactofilin HTH_8 AAA_lid_14 NtrC-AAA Receiver HTH_66 DUF433 PIN His-Kinase His-Kinase

NZ_DATKTK010000069.1:6705-21189
Sulfuricella sp.

Glycos_transf_1 Glyco_transf_4 Glyco_transf_9 Glycos_transf_1 Glyco_trans_1_2 Wzy_C Glyco_transf_9 MqsA_antitoxin MqsR_toxin CoA_binding_3 Bac_transf GDP_Man_Dehyd GDP_Man_Dehyd Epimerase Glyco_transf_4 Glycos_transf_1 RelE HTH_31

NZ_DATAJT010000631.1:1-4286
Vineibacter terrae

Bac_luciferase GFA MqsR_toxin MqsA_antitoxin Bac_luciferase Amidohydro_2